

Yincheng Zhou

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EDUCATION

University of Pennsylvania

Vagelos Integrated Program in Energy Research (VIPER)

Philadelphia, PA

September 2025 – Present

- **Dual Degree:** B.S.E. in Computer Science (AI Concentration) | B.A. in Physics & Astronomy
- **Relevant Coursework:** Big Data Analysis, Computer Systems, Data Structures & Algorithms, OOP
- **GPA:** 4.0 / 4.0

EXPERIENCE

GRASP Laboratory | University of Pennsylvania

Philadelphia, PA

Machine Learning Researcher, Advisor: Prof. Pratik Chaudhari

January 2026 – Present

- Proposed SPD, a sparse monocular depth estimator that predicts depth only at queried pixels, achieving $6\times$ inference speedup over dense methods while preserving accuracy (AbsRel 0.11 on NYU Depth V2) for real-time robotics.
- Developed a dense-train / sparse-infer paradigm with a 350K-param split decoder and cross-attention fusion on DINOv2 features, proving sparse output matches dense pixel-wise prediction through systematic ablation across 15 architecture variants.

Welta, Inc. | AI-powered cross-platform ad optimization engine [Demo]

Remote

Software Engineer Intern

March 2026 – Present

- Designed ad spend optimizer using Bayesian response curves and convex optimization to reallocate budget across channels at fixed cost; validated on 93 e-commerce brands, delivering 23% higher return on ad spend comparing to human baseline.
- Engineered ad simulation engine that tokenizes creatives and models persona-level reactions via LLM; reduced inference latency to 20% of baseline while improving prediction accuracy by 55%.

Human Machine Intelligence Laboratory | Peking University

Remote

Machine Learning Researcher, Advisor: Prof. Shanghang Zhang

March 2026 – Present

- Architected GUI agent post-training pipeline that generates step-level SFT data with MCTS and difficulty-weighted loss, outperforming agentic RL (GRPO) at 25% of compute; raised OSWorld success rate from 12.3% to 17.7% (+44% relative).
- Engineered distributed rollout infrastructure across 112 Docker VMs and 8 A100 GPUs with fault-tolerant container monitoring and action-replay VM cloning; collected 2,927 trajectories and curated 35K steps via 200 parallel LLM auditing agents.

Franklink, Inc. | World's first AI-native networking platform [Demo]

Philadelphia, PA

Founding Software Engineer

October 2025 – February 2026

- Designed iMessage-based multi-agent orchestration layer with persisted memory, selective dispatch across 5 sub-agents, and agentic RAG over 8 permissioned tools (DB, email, calendar); reduced user-facing latency by 30%.
- Engineered a Kafka-backed event pipeline with at-least-once delivery, idempotency keys, and retry + DLQ; sustained 50+ concurrent sessions with p99 latency within 40% of single-session baseline; deployed on AWS ECS.

PROJECTS

AlphaOne | Subject-oriented financial sentiment platform [GitHub] [Demo] [HuggingFace]

August 2025 – February 2026

- Fine-tuned DeBERTa for per-ticker aspect-based sentiment in multi-entity financial text, achieving 82.5% accuracy (+65pp over FinBERT); curated 6,287 training triples via Reddit scraping, LLM labeling, and manual audit of 4,501 pairs.
- Engineered a 3-pass batch NLP pipeline (split, infer, commit) with claim-based row locking for concurrent-safe ingestion across 7 subreddits; serves real-time per-ticker trends through a full-stack dashboard.

F-Zero Racing Agent | Reinforcement learning for competitive SNES racing [GitHub]

March 2026 – April 2026

- Trained a PPO agent across 80 parallel SNES emulators with custom reward shaping to race F-Zero, reaching global top 300 on a task previously deemed unsolvable due to sparse terminal-only rewards over 3,000+ step horizons.
- Engineered dense reward from raw RAM by spline-interpolating 58 checkpoints into a 660-point track centerline; designed dual-input CNN+MLP policy fusing 4-frame visual input with 59-dim state features.

IncidentGuard | Predict-then-act multi-agent incident remediation [GitHub]

April 2026

- Built a multi-agent system that predicts remediation outcomes before execution, rejecting dangerous reflexes via counterfactual ranking; reduced simulated recovery latency by 46% over naive baseline across 4 failure scenarios.
- Implemented closed-loop plan-execute-verify pipeline with live Prometheus metrics, OpenClaw browser automation, and DuckDB case persistence; system gracefully degrades across 3 fallback paths when infrastructure is unavailable.

NL2SQL Bot | Natural language to SQL with agentic orchestration [GitHub] [Demo]

December 2025 – January 2026

- Orchestrated 5 specialized agents on Google ADK with loop-guarded retry, per-agent model routing, and automated chart generation across 6 visualization types from natural language queries.
- Enforced read-only execution via 18 blocked SQL patterns and table allowlisting; abstracted dialect-specific rules across MySQL, PostgreSQL, and SQLite for cross-database portability.

SKILLS & ACHIEVEMENTS

Languages: C++, Python, Java, JavaScript/TypeScript, SQL, Assembly, OCaml

Backend & Infrastructure: FastAPI, Spring Boot, Kafka, Redis, Celery, Docker, AWS, Ray, Git

ML & Data: PyTorch, CUDA, vLLM, HuggingFace Transformers, spaCy, Pandas, LangGraph

Achievements: Team China, International Science and Engineering Fair (ISEF); Silver Medal, Team China, International Young Physicists' Tournament (IYPT); Second Prize, China Mathematics Olympiad; Second Prize, China Physics Olympiad; Champion, Start-up in a Weekend Hackathon, Photon Hybrid Intelligence Track